

QA LEADERSHIP ACADEMY

AI QA Use-Case Matrix

Plot candidate AI uses in QA on value against risk so you pilot the right ones in the right order.

Purpose

This matrix helps a QA leader decide where to apply AI first by plotting each candidate use case on two axes: potential value and residual risk. It replaces scattered, informal experimentation with a deliberate portfolio view, so you start with high-value, lower-risk uses and put guardrails around the tempting-but-risky ones before anyone touches them.

When to use it

- When AI use is already happening informally and you need to bring it into the open and prioritise it.
- When leadership asks "how should we use AI in QA?" and you want a structured answer.
- Before committing to any AI pilot, to confirm it sits in the value/risk zone you intend.

Instructions

1. List candidate use cases specifically ("draft test ideas from an acceptance criterion", not "use AI").
2. Score each for value (1-5) — time saved or quality gained — and for risk (1-5) — considering hallucination, data exposure and over-reliance.
3. Place each in one of four quadrants using the scores.
4. Sequence: do the high-value / low-risk quadrant first; gate the high-value / high-risk quadrant behind controls; ignore low-value / high-risk.
5. For anything you take forward, hand it to the AI Risk Assessment and AI Pilot Template before use.

Worked example

Candidate uses for Northstar, where a couple of testers already paste code and tickets into public AI tools with no policy:

Use case	Value (1-5)	Risk (1-5)	Quadrant	Action
Draft test ideas / scenarios from acceptance criteria	4	2	High value / low risk	Adopt first, with human review
Summarise a defect thread into a clear report	3	2	High value / low risk	Adopt, no sensitive data pasted
Suggest boundary/edge cases for a documented rule	4	2	High value / low risk	Adopt, human decides what to keep
Generate automation code from app source	4	4	High value / high risk	Pilot only behind IP/data controls
Analyse production data / logs with PII	3	5	High value / high risk	Do not, until data controls agreed
Auto-generate then auto-merge tests unreviewed	2	5	Low value / high risk	Reject — no human in the loop
Draft release notes from merged tickets	2	2	Low value / low risk	Optional, low priority

The immediate move for Northstar is to legitimise the high-value / low-risk uses (test-idea generation, defect summarisation) with a simple rule of "no source code or customer data pasted into public tools", which channels the existing informal behaviour into a safe lane. The tempting "generate automation from source" sits in high-value / high-risk and must go through a controlled pilot, not ad-hoc pasting into a public tool.

SEQUENCE BY QUADRANT

Adopt high-value/low-risk now, pilot high-value/high-risk behind controls, and simply reject low-value/high-risk. The quadrant, not the enthusiasm, sets the order.

Blank reusable version

Use case	Value (1-5)	Risk (1-5)	Quadrant	Action

Quadrant guide

- High value / low risk — adopt now with human review.
- High value / high risk — pilot only behind explicit controls (data, IP, human-in-the-loop).
- Low value / low risk — optional, low priority.
- Low value / high risk — reject.

Common mistakes

COMMON MISTAKES

Scoring "use AI" as one lump instead of specific use cases; underrating risk because a tool is easy and impressive; starting with the exciting high-risk quadrant before any controls exist; and treating the matrix as

permission to proceed — it prioritises, but the Risk Assessment and Pilot Template are what actually authorise a use.

How to interpret the results

- A dense high-value / high-risk quadrant means your first job is controls, not adoption.
- If informal use is already clustering in a risky quadrant, that is an urgent governance signal, not a productivity win.
- Value scores should reflect value after human review — AI that needs full re-checking may save less than it appears.
- Re-plot as controls mature: a high-risk use can move to adoptable once data and IP guardrails exist.

INSIDE STLC PRO TIP

When AI use is already happening under the radar — as at Northstar — do not open with a ban. Open with this matrix: name the safe, high-value uses your people are already reaching for, bless those officially, and you earn the credibility to put real guardrails around the risky ones.